

Late Breaking Results: FPGen-3D: Automated Framework for 3D-FPGA Architecture Generation and Exploration

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Abstract—In this work, we propose FPGen-3D, an automated framework for 3D field-programmable gate arrays (FPGA) architecture generation and exploration. FPGen-3D generates custom 3D FPGA fabrics based on user-defined architectural parameters, producing synthesizable register-transfer level (RTL) code, routing results, and programming bitstreams, enabling efficient navigation of the 3D FPGA design space. We demonstrate its capabilities through a case study of generating and exploring various vertical connection strategies via 3D switch blocks (SB) and 3D configurable logic block (CLB). Results show that 3D FPGAs can achieve lower wire length than traditional 2D FPGAs with a decrease of up to 18.5%.

I. INTRODUCTION

Recent advancements in packaging technologies have enabled the integration of 3D integrated circuits, offering significant improvements in performance, power, and area (PPA). As FPGAs continue to evolve, the adoption of 3D architectures presents new opportunities for achieving higher interconnect densities and reduced signal delays. However, designing and optimizing 3D FPGAs poses a substantial challenge due to their vast and unexplored design space, necessitating specialized tools capable of navigating the many architectural trade-offs.

Related Work. Several efforts have been made in 3D FPGA design. Amagasaki et al. proposed a 3D FPGA architecture featuring 3D SBs achieving a 5.3% reduction in delay compared to 2D designs [1]. Turkyilmaz et al. suggested a design where the configuration memory is placed on one layer, while the logic blocks occupy another, resulting in a 14% improvement in delay over 2D [2]. Chen et al. introduced an architecture with logic blocks on one layer and routing and BRAM on another which demonstrated a 54.9% reduction in wire length [3]. Boutros et al. made the output signals of logic blocks 3D, showing a wire length reduction of 4-5% [4].

Despite these efforts, a major limitation is their focus on single or a few fixed FPGA architectures, missing the vast 3D FPGA design space. There is currently no efficient method for automating the fabrication and testing of arbitrary 3D FPGA architectures, leaving much of the design space unexplored. **Our Contribution.** To address this gap, we propose FPGen-3D, an automated 3D FPGA architecture generation and exploration tool. FPGen-3D takes in user inputs to specify the 3D FPGA architecture, generates RTL code which can be synthesized, executes routing for FPGA benchmarks, and generates configuration bitstreams. By automating arbitrary 3D FPGA architecture generation, FPGen-3D enables efficient evaluation of various design parameters, provides insights into the advantages and disadvantages of different 3D FPGA implementations, and allows the mass exploration of the 3D FPGA design space.

II. FPGEN-3D FRAMEWORK

FPGen-3D is capable of: (1) automatic 3D FPGA RTL generation; (2) evaluation of user-defined 3D FPGA architectures; (3) placement and routing with 3D SBs; (4) 3D SB connection pattern and location exploration; (5) 3D FPGA bitstream generation. FPGen-3D also has the potential for accurate 3D FPGA PPA evaluation through physical design (work-in-progress).

A. Front-end: 3D FPGA Architecture Generation

FPGen-3D front-end accepts architectural parameters and benchmark circuits as input and produces the FPGA fabric RTL code. The generated RTL can be synthesized and physically placed and routed by physical design tools to obtain accurate information about the fabric, including thermal properties, power consumption, performance metrics, and area utilization (beyond the current scope of this framework; work-in-progress). Table I lists the different architectural parameters that are currently supported by FPGen-3D. FPGen-3D supports both homogeneous and heterogeneous 3D architectures. In heterogeneous configurations, routing units, CLBs, DSPs, and BRAMs can be distributed across layers according to user specifications. While placement and routing for some configurations are still in progress, RTL generation is available for any specified configuration.

FPGA Parameters	Options
Number of Layers	1, 2, 3, ...
Type of Layers	Homogeneous, Heterogeneous
Vertical Connection Type	Outputs only, Inputs and Outputs, 3D Switch Blocks
Switch Block Vertical Pattern	Wilton, Subset
Percentage of SBs that are 3D	0 to 100%
3D SB Locations	Perimeter, Center, Random, Repeated Interval
Normal FPGA parameters	Channel Width, Grid Size, LUT size, IO Fan-out, etc.

TABLE I: FPGen-3D Architecture Parameters Options

FPGen-3D leverages a modified version of OpenFPGA [5], an open-source fabric and bitstream generator for FPGAs. OpenFPGA has been extensively modified to support the generation of fabric RTLs and bitstreams for various 3D FPGAs architectures. Our key modifications include the addition of vertical routing channels and new sides to the SBs (Above and Under) to accommodate vertical connections. Connection blocks (CBs) and CLBs have been adjusted to be able to send signals both vertically and horizontally, depending on the architectural requirements. OpenFPGA was also adapted to recognize these new vertical connections possibilities and the concept of a multi-layer fabric.

B. Back-end: Placement, Routing, Bitstream Generation

After generating RTL for 3D FPGA fabric, FPGen-3D utilizes the Verilog-to-Routing (VTR) framework [4] for the placement and routing of benchmarks onto the fabric. VTR supports some limited 3D FPGA architectures, only the ones where either the input or output signals of grid locations are vertical. According to [6], VTR can also support 3D SB architectures, but only a subset connection pattern is supported, with no option to modify the ratio of 3D to 2D SBs. Additionally, there is a lack of clear documentation on how to utilize this functionality, and our attempts to use it resulted in unroutable or incorrect FPGA fabrics.

To enable VTR to operate with more 3D FPGA architectures, FPGen-3D generates a custom Routing Resource Graph (RRG) that represents the 3D interconnect topology of the user-defined FPGA architecture. The new 3D RRG is created by using a base RRG with no vertical connections and then adding the necessary nodes and edges required to represent the desired 3D FPGA architecture. VTR then uses this custom 3D RRG to place and route.

III. EXPERIMENTAL RESULTS

A. Tool Showcase

To demonstrate the capabilities of FPGen-3D, a 3D FPGA fabric was generated, synthesized, and implemented onto an FPGA. Fig. 2 presents the Vivado floorplan obtained after implementation, illustrating that the generated RTL code by FPGen-3D is correct and synthesizable.

The fabric was then functionally verified by programming it using generated bitstream for the s298 MCNC benchmark in RTL simulation and comparing its output to the expected results. This process confirms the functional correctness of the generated design, and the validity of the bitstream generation. Subsequently, the RTL can now be advanced through the physical design process to accurately determine the PPA metrics of the design.

B. 3D Vertical Connection Study

To showcase the potential applications of FPGen-3D, a study was conducted to evaluate the effectiveness of different vertical connections in 3D FPGAs. The architectures utilized in the study are described in Table II, and Fig. 3 illustrates the interlayer connections present in these different 3D architectures. The study was designed such that all architectures have approximately the same number of CLBs. The study focuses on three types of vertical connections:

- 1) **3D Inputs and Outputs (Fig. 3a):** Input and output signals of grid locations are vertical, a configuration currently supported by VTR.
- 2) **3D Outputs (Fig. 3b):** Only output signals of grid locations are vertical, also supported by VTR and inspired by [4].

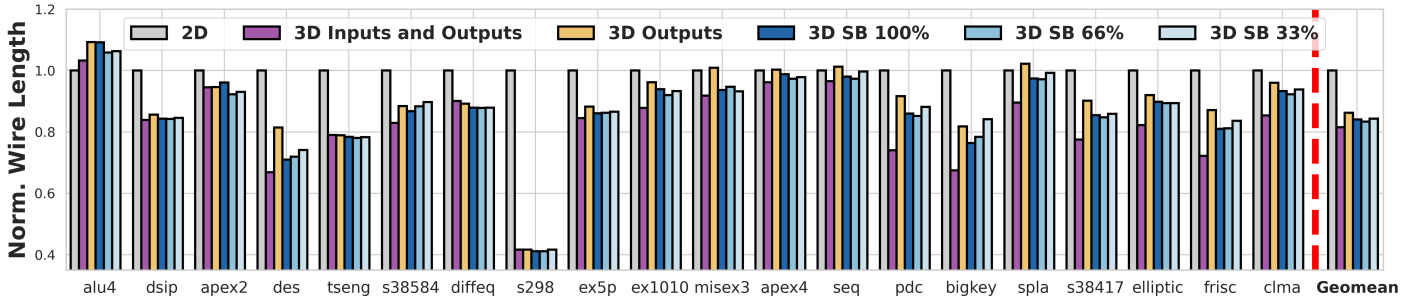


Fig. 1: Wire Length of the top 20 MCNC benchmarks on various 3D FPGA architectures normalized to the 2D FPGA architecture



Fig. 2: Implementation of 3D FPGA RTL onto a Xilinx Virtex Ultrascale+

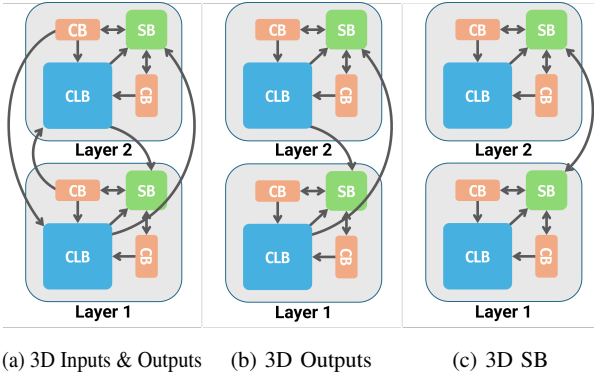


Fig. 3: Various 3D FPGA architectures supported by FPGen-3D

- 3) **Variable Percentage of 3D SBs (Fig. 3c):** We propose this design to explore the effects of varying the percentage of 3D SBs, with configurations of 1/3 (one out of every three SBs being 3D), 2/3 (two out of every two SBs being 3D), and all the Switch Blocks being 3D.

The different architectures were evaluated by placing and routing the 20 largest MCNC benchmarks [7] onto them and the routing results were analyzed. Since no specific physical design process was applied, accurate calculations of delay and area were not feasible. Therefore, the metric used for comparison was the total wire length required for routing on each architecture. This metric represents the total Manhattan distance traversed during a benchmark. **Takeaway 1: 3D architectures in general outperform 2D.** Fig. 1 presents the wirelength comparison for the 20 benchmarks, where Table III summarizes the normalized geomean values. It can be observed that all 3D architectures result in a lower wire length on average compared to the 2D architecture. The 3D Inputs and Outputs architecture performs the best, achieving an average reduction in wirelength of 18.5% ($0.815\times$ that of the 2D architecture). The 3D Outputs architecture shows the least

FPGA Parameter	Value
Grid Size	25x25 (3D), 35x35 (2D)
Channel Width	150
Number of Layers	2
Layers Type	Homogeneous
LUT Input Size	6
Number of LUTs per CLB	10
Wire Segment Length	1
Type of Vertical Connection	[Inputs and Outputs, Outputs only, (33%, 66%, 100%) 3D SB]
3D SB Placement Strategy	Repeated Interval
SB Connection Pattern	Wilton

TABLE II: Study 3D FPGA Architecture Parameters

3D FPGA Arch. Type	# of Vert. Connections	WL Geomean (Norm. to 2D)
3D Inputs and Outputs	340,170	4,383.59 (0.815)
3D Outputs	12,052	4,638.79 (0.863)
3D SB 100%	86,400	4,520.71 (0.841)
3D SB 66%	57,000	4,484.45 (0.834)
3D SB 33%	28,500	4,533.5 (0.843)

TABLE III: 3D FPGA Architecture Number of Vertical Connections and Wire Length Geometric Mean on MCNC benchmarks reduction, with an average wire length reduction of 13.7% ($0.863\times$). Our proposed 3D SB architectures, reduces wirelength by 15.9% to 16.6%.

These results indicate that while all 3D architectures offer improvements over the 2D design, the differences between them are relatively small. **Takeaway 2: 3D SBs designs show similar wire length reduction but with significantly fewer vertical connections.** Another major consideration for 3D FPGAs is the number of vertical connections present in the fabric. Since these vertical connections, depending on their type (e.g., through-silicon vias (TSVs), microbumps, monolithic interconnects (MIVs), etc.), can occupy significant space and impact fabrication complexity and cost, it is important to minimize the number of connections while seeking the best performance. As shown in Table III, the 3D Inputs and Outputs architecture has $3.9\times$ more vertical connections compared to the 100% 3D SB design and over $28\times$ more than the 3D Outputs design, but only a marginal benefit in wirelength (2-3%). The 3D SB architectures, on the other hand, outperforms the 3D Outputs design while having fewer vertical connections than the 3D Inputs and Outputs architecture, making them possibly an attractive choice with good tradeoffs.

Further investigation is needed to understand the effects of varying the percentage of 3D SBs, as the 66% version unexpectedly outperforms the 100% version despite having fewer vertical connections. This shows that other factors, e.g., place and route algorithms, may influence the performance and require additional exploration.

CONCLUSION

In this work, we introduced FPGen-3D, an automated tool that streamlines the design and exploration of 3D FPGA architectures. By generating custom 3D FPGA fabrics based on user-defined parameters and benchmarks, FPGen-3D facilitates efficient exploration of the vast 3D FPGA design space. Our results demonstrate that 3D FPGA designs can potentially achieve improved performance metrics compared to traditional 2D FPGAs.

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